

AgriSpike™ LoRa Field Telemetry

Decoded Packet Breakdown & System Diagnostics

TARGET NODE: NODE 1 PROTOCOL: LORA COMPACT KEY:VALUE TRANSMISSION STATUS: DECODED

Overall Node Status: Healthy

Node 1 is fully operational across all environmental and system parameters. Optimal soil moisture levels, zero crop heat stress, robust solar charging, and no system fault codes detected.

⚠ Event Flag Detected (PIR: 1)

The passive infrared sensor recorded local motion near the AgriSpike unit (indicating potential human or animal activity in proximity to the node).

Decoded Telemetry Packet Data

KEY	PARAMETER	VALUE	INTERPRETATION & RANGE ANALYSIS
ID	Node ID	1	Data received from Node 1
AT	Air Temperature	28.6 °C	Normal Standard ambient range
AH	Air Humidity	67.2 %	Healthy Favorable relative humidity
ST	Soil Temperature	25.8 °C	Normal Root zone temperature stable
SM	Soil Moisture	58 %	Optimal Target band (30%–70%)
LDR	Light / Sunlight Level	82 %	Bright daytime / strong solar radiation
DAY	Day Flag	1	Daytime cycle active (1 = Day, 0 = Night)
DL	Daylight Duration	11.6 hrs	Accumulated daylight hours since sunrise
ET	Evapotranspiration Index	63.4 / 100	Moderately high water loss rate from soil/plant
BV	Battery Voltage	4.05 V	Healthy Li-ion cell operating voltage
BP	Battery Percent	81 %	Good charge state
SV	Solar Panel Voltage	5.78 V	Solar panel actively charging power pack
SS	Soil Sensor Status	ACTIVE	Soil probe normal (not disconnected or stuck)
IRR	Irrigation Required	0	No Action Moisture adequate; no irrigation needed
HS	Heat Stress Flag	0	Normal Crop within safe thermal limits
PSI	Plant Stress Index	24 / 100	Low Stress Plant health conditions optimal
SYS	System Status	OK	No hardware or protocol faults detected
MODE	Power Mode	NORMAL	Standard operating mode (power-save inactive)
ERR	Error Code	0	No sensor read or transmission errors
PIR	Motion / Animal Sensor	1	Alert Motion detected in field proximity

Firmware Field Decoder Legend

Raw over-the-air LoRa packets transmit enumerations as compact integers to optimize radio bandwidth. Below is the mapping layer used by the AgriSpike dashboard:

FIELD KEY	RAW RADIO VALUE	DASHBOARD STRING	MEANING
SS (Soil Status)	0 / 1 / 2	ACTIVE / ERR / OFF	0 = Normal probe reading
SYS (System Status)	0 / 1 / 2	OK / WARN / FAIL	0 = Operational healthy state
MODE (Power Mode)	0 / 1 / 2	NORMAL / ECO / SLEEP	0 = Full operational mode